ITA06020 MACHINE LEARNING

Experiment 5 – Implementation of K-Nearest Neighbours (K-NN)

To implement the K-Nearest Neighbours (K-NN) algorithm in Python to classify data based on the nearest neighbouring **Aim:**

samples.

Dataset: Wine Recognition Dataset (real-world benchmark dataset available in Scikit-learn).

Procedure

1. Import the required Python libraries.
2. Load the Wine Recognition dataset.
3. Split the dataset into training and testing data.
4. Normalize the dataset using StandardScaler.
5. Create a KNN classifier with K = 5.
6. Train the model using the training data.
7. Predict the output using testing data.
8. Calculate and display the accuracy.

Python Program

```
from sklearn.datasets import load_wine

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score

data = load_wine()

X = data.data

y = data.target

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)

model = KNeighborsClassifier(n_neighbors=5)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("KNN Classification")

print("Training Samples:", len(X_train))

print("Testing Samples:", len(X_test))

print("Predicted Output:", data.target_names[y_pred[0]])

print("Actual Output:", data.target_names[y_test[0]])

print("Accuracy:", round(accuracy_score(y_test, y_pred)*100, 2), "%")
```

Output

```
KNN Classification
Training Samples: 142
Testing Samples: 36
Predicted Output: class_0
Actual Output: class_0
Accuracy: 94.44 %
```

Result:

Thus, the K-Nearest Neighbours algorithm was successfully implemented using Python. The model achieved an accuracy of 94.44% on the Wine Recognition dataset.